

2.1

	E_{in}	E_{x1}	E_{y1}	E_{x2}	E_{y2}	E_r	E_t
E_{in}	0	$t_{BS} e^{-i\phi_x}$	$r_{BS} e^{-i\phi_y}$	0	0	0	0
E_{x1}	0	0	0	$r_x e^{-i\phi_x}$	0	0	0
E_{y1}	0	0	0	0	$r_y e^{-i\phi_y}$	0	0
E_{x2}	0	0	0	0	0	t_{BS}	$-r_{BS}$
E_{y2}	0	0	0	0	0	r_{BS}	t_{BS}
E_r	0	0	0	0	0	0	0
E_t	0	0	0	0	0	0	0

using $\phi_d = \omega_0(L_x - L_y)/c$

matrix results in

$$E_{AS} = E_c + E_+ + E_-$$
$$= E_0 (e^{-i\omega_0 t} \sin \phi_d + [e^{-i(\omega_0 + \Omega)t} \sin(\phi_d + \phi_\Omega) + e^{-i(\omega_0 - \Omega)t} \sin(\phi_d - \phi_\Omega)])$$

so the length terms are balled into the phase terms

when $L_x = L_y$

$$\phi_d = \phi_\Omega = 0$$

so,

$$E_{AS} = E_0(0 + 0 + 0) = 0$$

ie dark fringe

2.2

$$P_{AS}(t) = |E_{AS}(t)|^2$$
$$= E_0^2 (e^{-i\omega_0 t} \sin \phi_d + [e^{-i(\omega_0 + \Omega)t} \sin(\phi_d + \phi_\Omega) + e^{-i(\omega_0 - \Omega)t} \sin(\phi_d - \phi_\Omega)] (e^{i\omega_0 t} \sin(\phi_d + \dots)))$$
$$= E_0^2 (\sin^2 \phi_d + 2 \Gamma \sin \phi_d [\underbrace{\sin(\phi_d + \phi_\Omega) \cos(\Omega t)}_{2 \sin(\phi_d) \cos(\phi_\Omega)} + \underbrace{\sin(\phi_d - \phi_\Omega) \cos(\Omega t)}_{2 \sin(\phi_d) \cos(\phi_\Omega)})]$$

* using $\Gamma^2 = 0$

$$P_{AS}(t) = P_0 (\sin^2 \phi_d + 4 \Gamma \sin^2 \phi_d \cos \phi_\Omega \cos(\Omega t))$$

2.3

i am confused, plot was not a PDH error signal

3.1

	E_{in}	E_{x1}	E_{y1}	E_{x2}	E_{y2}	E_{x3}	E_{y3}	E_r	E_t
E_{in}	0	$t_{BS} e^{-i\phi_x}$	$r_{BS} e^{-i\phi_y}$	0	0	0	0	0	0
E_{x1}	0	0	0	$t_{TMX} e^{-i\phi_x}$	0	0	$-r_{TMX} e^{-i\phi_x}$	0	0
E_{y1}	0	0	0	0	$t_{TMX} e^{-i\phi_y}$	0	$-r_{TMX} e^{-i\phi_y}$	0	0
E_{x2}	0	0	0	0	$r_{ETMX} e^{-i\phi_x}$	0	0	0	0
E_{y2}	0	0	0	$r_{ETMX} e^{-i\phi_y}$	0	0	$t_{ETMX} e^{-i\phi_y}$	0	0
E_{x3}	0	0	0	0	0	$r_{ETMY} e^{-i\phi_x}$	0	0	0
E_{y3}	0	0	0	0	0	$r_{ETMY} e^{-i\phi_y}$	0	0	0
E_{r3}	0	0	0	0	0	0	0	t_{BS}	$-r_{BS}$

$$\begin{array}{cccccccccccc}
 E_{vr} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & r_{bs} & t_{bs} \\
 E_r & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 E_r & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{array}$$

3.2

$$\frac{E_{as}}{E_{in}} = \frac{1}{2} \left[e^{-2i\phi_x} \left(r_{\pi\pi x} - \frac{t_{\pi\pi x} r_{e\pi x} e^{-2i\phi_x}}{1 - r_{\pi\pi x} r_{e\pi x} e^{2i\phi_x}} \right) - e^{-2i\phi_y} \left(r_{\pi\pi y} - \frac{t_{\pi\pi y} r_{e\pi y} e^{-2i\phi_y}}{1 - r_{\pi\pi y} r_{e\pi y} e^{2i\phi_y}} \right) \right]$$

3.3

with simplifications you get

$$\frac{E_{as}}{E_{in}} = \frac{1}{2} [r_{fp}(\Phi_c + \Phi_d) - r_{fp}(\Phi_c - \Phi_d)]$$